

MedVault - Smart Digital Health Record and Appointment System

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Introduction

- MedVault is a modern web-based Digital Health Ecosystem designed to connect Patients, Doctors, Receptionists, Administrators, and Pharmacists into a single platform.
- Replaces traditional paper-based prescriptions and fragmented manual OPD register entries with structured digital workflows.
- Features dedicated role-based portals for Patients (vitals & EMR vault), Doctors (consultation queue & digital Rx builder), and Receptionists (walk-in token management).
- Built using React 18, TypeScript, and Tailwind CSS on the frontend, powered by Supabase PostgreSQL for secure role-based data access.
- Goal: Create a lightweight, high-contrast, zero-delay healthcare portal that digitizes medical records and streamlines hospital consultations.

Background and Motivation

- **Loss of Paper Prescriptions & Reports:** Physical medical slips are easily misplaced or damaged, leaving doctors without patient medical history during follow-ups.
- **Uncoordinated Reception & OPD Queues:** Walk-in patients face long waiting lines without real-time updates between reception desks and doctor consultation rooms.
- **Illegible Handwriting & Medication Risks:** Handwritten prescriptions increase the risk of dosage confusion for pharmacists and patients.
- **Lack of Centralized Digital Vaults:** Patients struggle to access their lab diagnostic reports, vitals telemetry history, and active medication lists in one place.
- **Motivation:** Build an easy-to-use, high-performance web system that eliminates paper delays, automates OPD queue pushing, and generates verified digital A4 prescriptions.

Relevance and Importance

- Instant Consultation Queue Sync: Tokens created at reception immediately appear on the doctor's live queue without manual paper slips or page refreshes.
- Verified Digital Prescriptions: Doctors generate clean A4 digital prescriptions with dosage frequency (1-0-1), doctor signature stamp, and scannable QR verification.
- Patient Vitals Telemetry Tracking: Allows patients to track Blood Pressure, Heart Rate, SpO2, and Blood Glucose history in visual charts.
- Data Security & Row-Level Access: Ensures strict patient privacy where users only access their authorized medical records using Supabase RLS security policies.
- Scalable Architecture: Designed as an extensible foundation ready for government ABHA ID integration, video call telehealth, and hospital bed analytics.

Literature Review

Sr.No	Ref.No(Citation)	Algorithm / Approach	Dataset / Source	Key Result	Findings & Limitations
1	[1]	Commercial Appointment Booking (Practo / Apollo 24/7)	Closed Commercial Database	Doctor search and appointment booking	Closed-source, subscription-based; lacks integrated reception token queue or doctor Rx generation.
2	[2]	Legacy Hospital Information Systems (HIS)	Institutional Desktop Software	Internal hospital record management	Heavyweight desktop software; no patient-facing portal, digital health locker, or mobile access.
3	[3]	Standalone EMR Storage Architectures	Clinical EHR Datasets	Centralized medical record storage	Focuses only on static storage; lacks live consultation line sync, digital signatures, or QR verification.
4	[4]	Proposed MedVault Ecosystem	Supabase Relational Database	Integrated 5-role digital healthcare platform	Lightweight, open web architecture combining live reception queue push, digital Rx PDF export, and vitals telemetry.

Research Gap

- Commercial Platforms Lack Hospital-Side Integration: Platforms like Practo or Apollo 24/7 handle online booking but do not provide reception token generators or live doctor consultation queues.
- Legacy HIS Systems Are Heavyweight & Complex: Traditional hospital software requires expensive local server setups and lacks clean, responsive patient-facing web interfaces.
- No Integrated Prescription Signature Engine: Existing web portals rarely combine diagnosis entry, dosage frequency (1-0-1), doctor digital stamp, and A4 PDF export in one workflow.
- Fragmented Vitals Monitoring: Patients lack a unified digital health locker where live vitals, past prescriptions, and appointment records are accessible in one account.
- Research Gap Solved by MedVault: MedVault fills this gap by delivering a unified, role-based web platform that seamlessly connects Receptionists, Doctors, and Patients.

Objectives

- Design a Role-Based Healthcare Web Application providing customized portals for Patients, Doctors, Receptionists, Administrators, and Pharmacists.
- Implement Real-Time Reception Token Sync to automatically push walk-in patient tokens from reception desks to doctor consultation screens.
- Develop a Digital Prescription Studio featuring medicine selection, dosage timing (1-0-1), doctor digital stamp, and A4 PDF export with QR codes.
- Build a Patient Health Vault for tracking Blood Pressure, Heart Rate, SpO2, and Blood Glucose with digital document storage.
- Enforce Database-Level Row Security (RLS) using Supabase PostgreSQL to protect sensitive patient records and maintain medical privacy.

Required Dataset, Tools and Technology

- **Frontend UI Framework:** React 18, TypeScript, Tailwind CSS v4, Lucide React Icons
- **State Management & Sync:** Zustand central state store for real-time consultation queue updates
- **Database & Security:** Supabase PostgreSQL database with Row Level Security (RLS) policies
- **PDF & QR Export Engine:** jsPDF and html2canvas for generating verified digital A4 prescription PDFs
- **Development IDE & Tools:** Visual Studio Code, Vite Build Tool, Git Version Control
- **Deployment (Planned):** Vercel / Render cloud deployment for high-availability access
- **Dataset:** Not Applicable — System operates on real-time data generated by logged-in Patients, Doctors, and Receptionists.

Methodology

- Requirement Analysis: Identified operational bottlenecks in hospital OPD queues, paper prescriptions, and patient record tracking.
- System Architecture Design: Modularized the application into 5 interconnected role-based portals using React components.
- Database Modeling: Designed normalized PostgreSQL database tables for Users, Patients, Doctors, Appointments, and Prescriptions.
- State & Event Sync Implementation: Built Zustand store handlers to sync reception token entries directly with doctor consultation queues.
- Prescription Generator Development: Implemented A4 PDF rendering engine with digital doctor stamps and scannable QR verification.
- Testing & UI Refinement: Performed module-wise functional testing and enforced high-contrast accessible styling.

Result & Review 2 Status

- Patient Vault Completed: Live vitals dashboard (BP, HR, SpO2), digital health locker, and online appointment booking fully operational.
- Doctor OPD Studio Completed: Real-time patient queue list, digital prescription maker with 1-0-1 dosage options, doctor stamp, and PDF export fully working.
- Reception Desk Completed: Walk-in patient registration form, token creation, and live line synchronization working without page refreshes.
- Database & Security Verified: PostgreSQL schema deployed on Supabase with Row Level Security (RLS) policies protecting user records.
- Expected by Final Review: Complete integration of ABHA government ID lookup, emergency vitals triage, and WebRTC video call telemedicine.

Conclusion

- MedVault successfully bridges the gap between Patients, Doctors, and Receptionists by unifying healthcare operations into one modern web platform.
- Replaces unreadable paper prescriptions with verified digital A4 PDFs containing dosage guidance and scannable QR verification.
- Eliminates OPD queue delays through live reception-to-doctor token synchronization.
- Demonstrates a lightweight, high-performance web architecture that is easy to deploy, highly secure, and accessible on any device.

Future Work

- ABDM Government ABHA Health ID Integration: Connecting with national health gateways for 1-click cross-hospital EMR data decryption.
- Algorithmic Emergency Vitals Triage: Automated risk scoring (BP > 165/105 mmHg) to re-rank critical patients to Priority #1 in doctor queue.
- Native WebRTC P2P Telemedicine: Direct browser-to-browser encrypted video consultation with live canvas prescription drawing.
- Hospital Bed Occupancy & Inventory Analytics: Tracking available ICU/General beds and pharmacy stock levels for hospital administrators.

References

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